



Institutional Requirements for the Use of X-rays on Human Participants for Research Outcomes

Background

The University of Toronto (U of T) conducts a wide range of innovative research involving human participants. This document sets out the requirements for the use of X-rays and X-ray machines to conduct research involving human research participants at or under the auspices of U of T. These requirements take into consideration regulatory requirements, ethics considerations and maintaining radiation doses As Low As Reasonably Achievable (ALARA). The operation and registration of X-ray machines in Ontario is currently governed by the Healing Arts Radiation Protection (HARP) Act and the Occupational Health & Safety Act (OHSA). This document is issued pursuant to the University of Toronto [Health and Safety Policy](#) and the University of Toronto [Policy on Research Involving Human Subjects](#) to further compliance with those policies and the provisions of the HARP Act and OHSA.

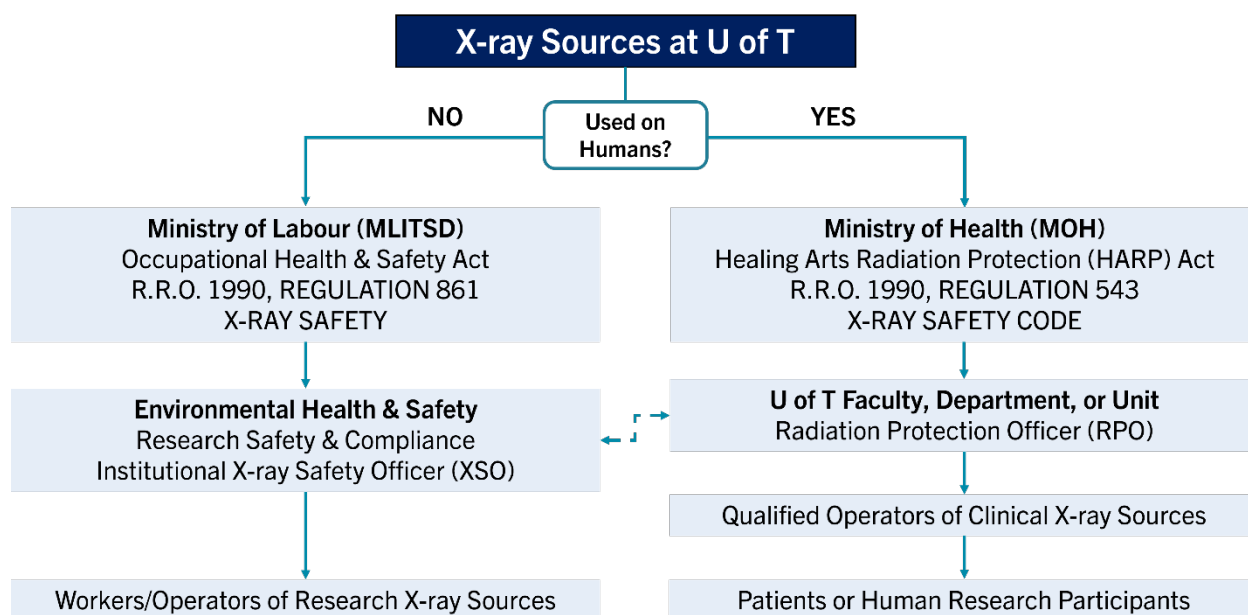


Figure 1: The regulatory framework for installing, operating, and maintaining oversight of X-ray machines at the University of Toronto. The Office of Environmental Health & Safety maintains oversight of all X-ray operators at the University.



Key Definitions

As Low As Reasonably Achievable (ALARA): a fundamental concept in radiation protection and safety that says exposure to ionizing radiation should be reduced to the lowest possible levels.

Research: is defined as an undertaking intended to extend knowledge through a disciplined inquiry and/or systematic investigation.

Human Participants: is defined as those individuals whose data, or responses to interventions, stimuli, or questions by the researcher, are relevant to answering the research question(s).

Research Ethics Board (REB): is an independent committee mandated to review the ethical acceptability of human research in accordance with federal, provincial and institutional requirements.

Ethics Protocol: is a document describing how the research will be conducted from a participant-centric perspective to ensure that all ethical requirements and standards are maintained.

Ionizing radiation: any type of radiation that has enough energy to ionize molecules in biological tissue

Equipment Owner: when used with reference to an X-ray machine, is the person designated by the University who has the management and control of the X-ray machine.

Permit Holder: a tenured or tenure-stream U of T faculty member that has the expertise, authority and resources to provide daily oversight over equipment and personnel under the permit.

Environmental Health & Safety (EHS): the office that is responsible for safe and healthy environment at the University. EHS is part of the Division of the Vice-President, Research and Innovation, and Strategic Initiatives.

Institutional X-ray Safety Officer (XSO): a person that is designated by the University that oversees all occupational matters related to the safe operation of X-ray machines at U of T and is a member of the Office of Environmental Health & Safety

Radiation Protection Officer (RPO): a person designated by the owner of the X-ray machine or facility that meets the qualifications under Section 9(1) and has the responsibilities under Section 4 of the HARP Act.

X-ray machine (HARP Act): an electrically powered device the purpose and function of which is the production of X-rays for the irradiation of a human being for a therapeutic or diagnostic purpose.

X-ray machine (OHSA): an electrically powered device, the principal purpose of which is the production of X-rays.

X-rays: artificially produced electromagnetic radiation with peak energy greater than five kilovolts.



1 Regulatory Framework at U of T

The use of X-ray machines at U of T currently falls under two main categories: non-clinical or occupational use, and clinical use. Non-clinical systems have a wide range of applications in non-biological samples and *in vivo* research with cell cultures or small animals. Clinical X-ray machines are those used for therapeutic and/or diagnostic treatments in patients as prescribed by a medical practitioner. This framework provides guidance for the operation of clinical X-ray machines on human research participants for non-medical purposes.

Non-clinical or occupational use of X-rays

X-ray machines that are used for research or non-clinical purposes are regulated under the Occupational Health and Safety Act (OHSA), which is enforced by the Ministry of Labour, Immigration, Training & Skills Development (MLITSD). The X-ray safety program developed under this legislation covers X-ray machines and personnel operating the machines (X-ray workers) at U of T. The X-ray safety program also applies to U of T workers operating clinical X-ray machines.

The University appoints a competent and qualified individual as the X-ray Safety Officer (XSO) to act on its behalf in matters related to the occupational use of X-ray machines with the Ontario MLITSD. The XSO is a member of the Office of Environmental Health & Safety (EHS) and is responsible for the implementation and enforcement of the U of T X-ray Safety Program.

Clinical use of X-rays

X-ray machines that are normally used clinically for therapeutic or diagnostic purposes (i.e., with patients) are subject to the HARP Act which is enforced by the Ontario Ministry of Health (MOH). Each faculty, department, or unit that uses clinical X-ray machines at U of T is responsible for their individual clinical X-ray safety program. The requirements under the HARP Act also apply to clinical X-ray machines used on human research participants.

The U of T faculty, department, or unit that owns the clinical X-ray machine is responsible for appointing a qualified and competent Radiation Protection Officer (RPO) that meets the requirements under Section 9 of the HARP Act. The RPO is responsible for obtaining approval for the installation and maintenance of clinical X-ray machines. Operators of X-ray machines used for this purpose must also meet specific qualifications under section 5 of the HARP Act. The institutional XSO is responsible for ensuring that operators of clinical systems are working safely in accordance with the OHSA.

2 Institutional Requirements

2.1 Approval by the University

Human Research Ethics Review & Approval

All researchers engaging in research activities involving human participants must obtain REB approval at U of T for their proposed project prior to commencing their work. Researchers wishing to



do so may submit their REB protocol through U of T's online ethics application system My Research Human Protocols (MRHP). The REB review ensures that the rights, safety, and welfare of human participants are protected. As part of the REB's review, they will pay close attention to the rationale/justification of the intervention (i.e., whether exposing participants to ionizing radiation is ethically acceptable). As part of the researchers' submission to the REB, they must include:

1. The proposed effective dose that will be delivered to each human participant per study, and any appropriate modifications to the dose based on the unique characteristics of the participant (e.g., age, weight, sex, etc.);
2. Inclusion and exclusion criteria for research participants;
3. The proposed duration of the exposure(s);
4. The proposed area of exposure of the human participant;
5. An ethically acceptable justification why the study requires exposure of human participants to ionizing radiation to address the research question;
6. Discussion on how risks and harms to participants will be minimized and mitigated; and
7. A plan to manage incidental findings, if reasonably foreseeable.

EHS Permit Application & Approval

To maintain appropriate institutional oversight and ensure regulatory compliance, clinical X-ray machines that are used for research activities involving human participants must be registered under an EHS X-ray Permit prior to commencing regular operations:

1. The Faculty/Department/Unit should identify an Equipment Owner that has sufficient authority to enforce the policies and responsibilities under the permit
2. The equipment owner is required to apply for an EHS X-ray Permit as the Permit Holder (PH) for the respective X-ray machine
3. The equipment owner and permit holder must be tenured or tenure-stream U of T faculty
4. Prior to commencing the permit application process, the equipment owner must ensure that the X-ray machine and facility have been registered and approved by the MOH (see Section 2.2).
5. Permit applications with supporting documentation must be submitted to EHS Radiation Protection Service for review and approval granted prior to beginning use of the equipment.

2.2 Approval by the Regulator(s)

U of T Faculties, Departments, or Units referred to as the "owner" of the clinical X-ray machines that intend on conducting research with human participants must obtain approval from the Ministry of Health (MOH) by fulfilling the following requirements:

1. Obtain approval from the MOH for the installation and registration of the clinical X-ray machine
2. Submit X-ray equipment and shielding specifications for each machine located in the designated facility to the MOH



3. Appoint a Radiation Protection Officer (RPO) that meets the qualifications listed in Section 9 of the HARP Act and is approved by the MOH
4. Ensure operators of clinical X-ray machines meet the qualifications under section 5 of the HARP Act, R.S.O. 1990, c. H.2, s. 5 (1).
5. Ensure clinical X-ray machines are installed, operated, and maintained according to provisions under the HARP Act, Regulation 543 and Health Canada Safety Code 35.

2.3 Operation of Clinical X-ray Machines

The “Equipment Owner” of the clinical X-ray machine(s) and the RPO must ensure that clinical X-ray machines are operated in accordance with all requirements outlined in this framework, as well as the following provisions under the HARP Act:

- a. No person shall operate an X-ray machine for the irradiation of a human being unless the person meets the qualifications and requirements prescribed by Section 5 of the HARP Act.
- b. No person shall operate an X-ray machine for the irradiation of a human being unless the irradiation has been prescribed by a qualified medical practitioner or another professional as prescribed by Section 6 of the HARP Act.

2.4 Personal Radiation Monitoring

Personal radiation monitoring of clinical X-ray operators is regulated by the Ontario MLITSD, Health Canada Safety Code 35 also sets recommendations for monitoring of personnel. At U of T monitoring of all personnel operating X-ray machines/sources is the responsibility of the XSO, who implements the radiation monitoring program through the Office of Environmental Health & Safety. The program includes quarterly dose monitoring using a federally certified service provider. The XSO individually evaluates the risk of exposure to ionizing radiation and recommends a monitoring regime according to the risk analysis.

2.5 Facility Design & Shielding

It is the responsibility of the U of T Facility RPO to ensure that the design of the X-ray facility that will house the clinical X-ray machine(s) meets the requirements under Regulation 543 of the HARP Act and Health Canada Safety Code 35. The RPO may engage the professional advice of the institutional XSO to assist in the assessment of the need and design for shielding of the X-ray facility.



3 Roles & Responsibilities

3.1 Faculty/Department/Unit and Equipment Owner

The Equipment Owner/Principal Investigator of a clinical X-ray machine has the following responsibilities within the institutional framework:

1. Submitting research proposals involving human research participants for REB review and approval
2. Applying for and obtaining approval from the MOH in accordance with Section 2.2, with support from EHS upon request.
3. Applying for and maintaining an EHS X-ray Safety Permit for the X-ray machine and facility
4. Participating in U of T EHS X-ray Safety program and annual EHS safety inspection program
5. Following and complying with the regulations under the Ontario HARP Act and requirements by the Ontario Ministry of Health
6. Ensuring qualified X-ray machine operators and facility occupants participate in EHS X-ray Safety training as well as providing job-specific training
7. Appointing a qualified Radiation Protection Officer (RPO)
8. Consenting research participants and ensuring they are made aware of the risks associated with exposure to ionizing radiation in addition to other study-related risks

3.2 Facility Radiation Protection Officer (RPO)

The Facility RPO is to be appointed by the responsible Faculty, Department, Unit or Equipment Owner of the clinical X-ray machine and approved by the X-ray Inspection Service (XRIS) of the MOH. The responsibilities of the RPO include but are not limited to:

1. Registering and maintaining registration of all applicable clinical X-ray machines and X-ray facilities with the Ontario Ministry of Health.
2. Establishing and maintaining procedures and quality tests for the safe operation of the X-ray equipment as per R.R.O. 1990, Regulation 543.
3. Ensuring that the X-ray equipment is maintained and meets the standards prescribed by the regulations as per R.R.O. 1990, Regulation 543.
4. Ensure acceptance and quality testing of the clinical X-ray machines is performed according to R.R.O. 1990, Regulation 543.
5. Ensuring that all X-ray operators are qualified according to the HARP Act and Health Canada Code 35 requirements.
6. Ensuring operations of an approved clinical X-ray machine and X-ray facility are in compliance with all regulations and requirements under the HARP Act, R.S.O. 1990, c. H.2.

3.3 Environmental Health & Safety (EHS)

The Office of Environmental Health & Safety and the institutional X-ray Safety Officer (XSO) have the following responsibilities:



1. Supporting researchers through the EHS X-ray permitting process and associated regulatory approvals.
2. Ensuring the Faculty/Department/Unit has appropriate approvals from the relevant regulatory bodies (i.e., MOH, MLITSD).
3. Ensuring qualified operators are working safely and in compliance with the regulations under the Ontario Occupational Safety and Health Act.
4. Providing dose monitoring services as required.
5. Conducting annual X-ray safety inspections as part of the broader U of T EHS X-ray Safety Program.

3.4 Research Ethics Board and Human Research Ethics Unit (HREU)

The Human Research Ethics Unit has the following responsibilities:

1. Supporting researchers in the development, review, and oversight of human ethics protocols to ensure compliance with applicable laws, regulations and policies.
2. Supporting researchers through the Research Ethics Board review and protocol submission process.
3. Providing one-on-one or group ethics consultations for faculty, staff and students prior to and throughout the ethics review process
4. Providing research ethics training and education for researchers

4 References

1. Occupational Health and Safety Act, R.R.O. 1990, Regulation 861, X-ray Safety; Ministry of Labour, Immigration, Training and Skills Development, Government of Ontario
2. Healing Arts Radiation Protection Act, R.S.O. 1990, Chapter H.2, Ministry of Health, Government of Ontario
3. Healing Arts Radiation Protection Act, R.R.O 1990, Regulation 543 – X-ray Safety Code, Ministry of Labour, Government of Ontario
4. Safety Code 35: Safety Procedures for the Installation, Use and Control of X-ray Equipment in Large Medical Radiological Facilities, Health Canada, June 2024
5. Canadian Institutes of Health Research, Natural Sciences and Engineering Research Council of Canada, and Social Sciences and Humanities Research Council of Canada, TriCouncil Policy Statement: Ethical Conduct for Research Involving Humans, December 2022.
6. Biomedical Research Involving Radiation Exposure, International Atomic Energy Agency (IAEA), <https://www.iaea.org/resources/rpop/health-professionals/nuclear-medicine/biomedical-research>